DRILL 08

HOSE & STREAMS

NFPA 1001 Level 1 (JPR 4.3.10) / WorkSafeNB Reg 91-191

EXTENDING A WORKING HOSELINE (BREAK & EXTEND)

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Firefighter Pairs

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must know how to extend an active, charged attack line that has come up short of the fire room without requiring the pump operator to shut down the entire discharge at the truck. Using the **Break-and-Extend** method or a manual **Hose Clamp / Z-Kink**, crews can add a 50-ft section in under 45 seconds right at the doorway.

📦 TOOLS & APPARATUS SETUP

- 1 charged 1.75" (45mm) attack line (100–150 ft) with combination or breakaway nozzle (shutoff ball valve with threaded 1.5" NST outlet behind tip).
- 1 spare 50-ft (15m) section of 1.75" hose rolled or flaked.
- 1 manual hose clamp (or training technique using double/Z-kink).
- Spare nozzle or original nozzle tip.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **High-Pressure Release:** When uncoupling a clamped or kinked hose, residual pressure can spray high-velocity water into faces. Ensure eye protection/visors are down.

⚠️ **Hose Clamp Pinch Points:** Keep fingers completely clear of mechanical clamp jaws and leverage levers.

🛑 **THE STOP RULE: Shout "STOP!" immediately if a kinked line slips loose or if high-pressure water whips uncoupled hose.**

PHASE 1: UNDERSTANDING THE BREAKAWAY NOZZLE & THE SHORT STRETCH (10 MINUTES) — SLOW DRY RUN

1. In rural residential structures (large farmhouses, split-levels, long driveways), 200 feet of preconnect sometimes falls 20–30 feet short of the second-story bedroom or attic.
2. If your department runs **breakaway nozzles**, the fog or smooth bore tip unscrews from the ball-shutoff valve, exposing a standard 1.5" male hose thread!
3. If not using a breakaway nozzle, the line must be clamped or z-kinked 10–15 feet behind the nozzle coupling.

DRILL 08: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Hose & Streams • Practical Application

PHASE 2: METHOD 1 – BREAKAWAY NOZZLE EXTENSION (15 MINUTES) — BUDDY COACHING

1. Nozzleman reaches maximum stretch: *"Line is short! Extending 50 feet!"*
2. Close the nozzle bail completely.
3. Unscrew the outer tip from the shutoff valve. The shutoff valve now acts as a closed gate valve holding back full line pressure.
4. Firefighter B connects the female coupling of a fresh 50-ft section directly onto the shutoff valve threads.
5. Stretch and flake the new 50-ft section into the structure.
6. Connect the nozzle tip (or a second nozzle) to the end of the new section.
7. Open the original shutoff valve slowly. Trapped air bleeds through the new nozzle; the line is now 50 feet longer and back in service!

PHASE 3: METHOD 2 – THE Z-KINK / HOSE CLAMP METHOD (20 MINUTES) — TIMED / STRESS

(Used when the nozzle is a one-piece unit without breakaway threads)

1. **The Z-Kink (Manual Clamp by Crew):**
   - Nozzleman and backup firefighter step 10 feet back from the nozzle.
   - Fold the hose back onto itself twice, creating a tight "Z" shape.
   - Both firefighters kneel onto the folds, putting their full body weight onto the kinks. The sharp folds choke off water pressure completely.
2. **Uncoupling & Connecting:**
   - A third firefighter (or nozzleman reaching forward) opens the nozzle bail to bleed pressure, then unscrews the nozzle from the coupling.
   - Attach the new 50-ft roll to the open coupling.
   - Attach the nozzle to the end of the new section.
   - On command: *"Releasing kink in 3, 2, 1!"* Firefighters slowly unbend the Z-kink, restoring full line pressure smoothly.
3. Practice with a mechanical **Hose Clamp** applied 15–20 feet back from the nozzle (always apply clamp with handle pointed toward the pump).

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Trying to unscrew a pressurized coupling:* Without bleeding line pressure through the nozzle bail first, the threads are locked by 100+ PSI of hydraulic force and will not turn.
- ❌ *Releasing clamp or kink too quickly:* Slamming water back into an empty 50-ft section creates a violent pressure spike and water hammer. Open the clamp or release the kink slowly!
- ❌ *Dropping the uncoupled nozzle in mud:* Dirt inside the nozzle tip clogs internal stream shapers and ruins fog patterns.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the Z-kink hold back pressure without slipping?", "How much faster is a breakaway nozzle extension compared to shutting down the pump panel?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Nozzle tips reattached and torqued hand-tight.
- ☐ Spare 50-ft section drained, rolled in donut or straight roll, and returned to storage.
- ☐ Hose clamp returned to apparatus bracket.
- ☐ Gaskets checked in all broken couplings.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS